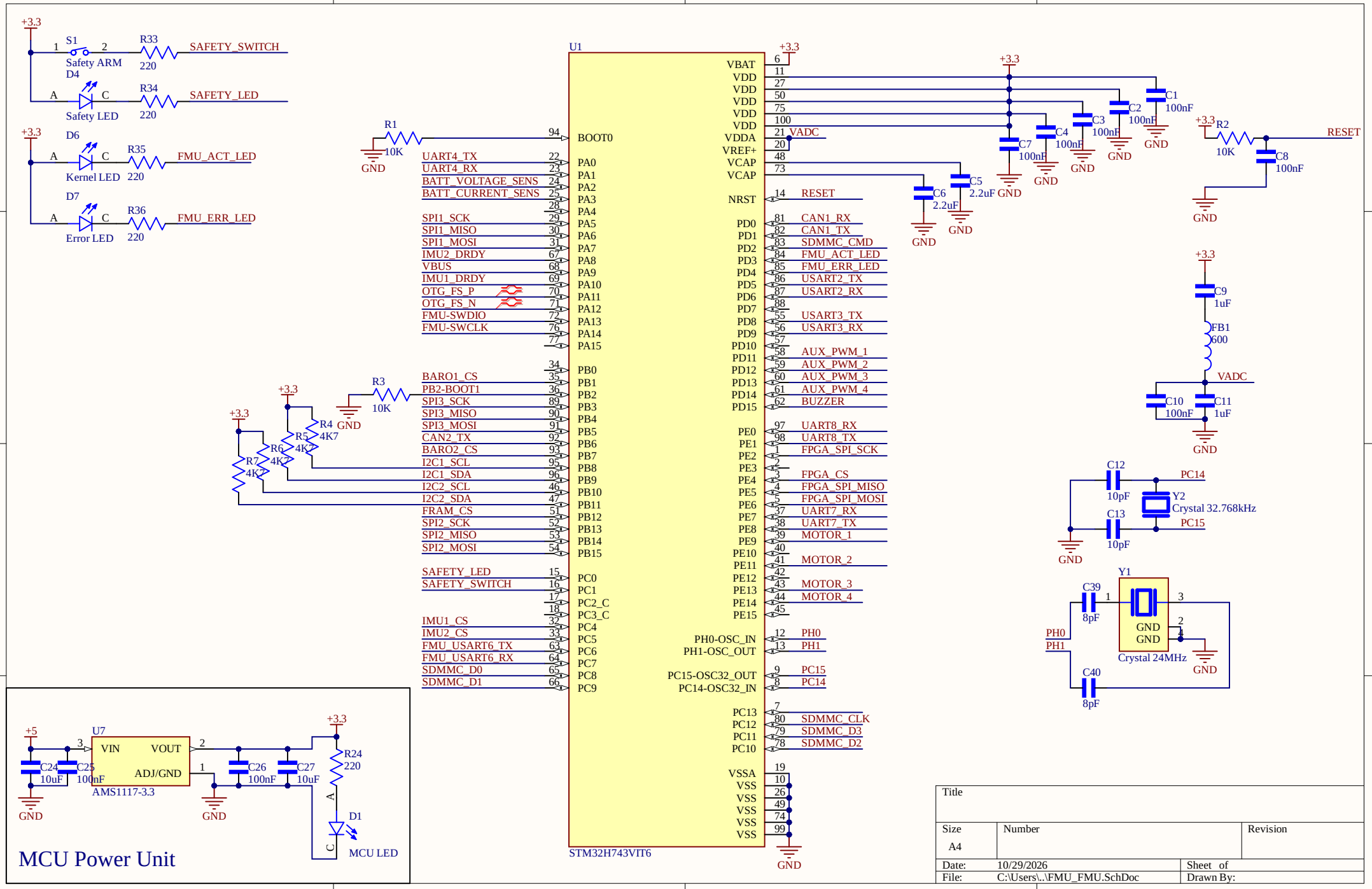
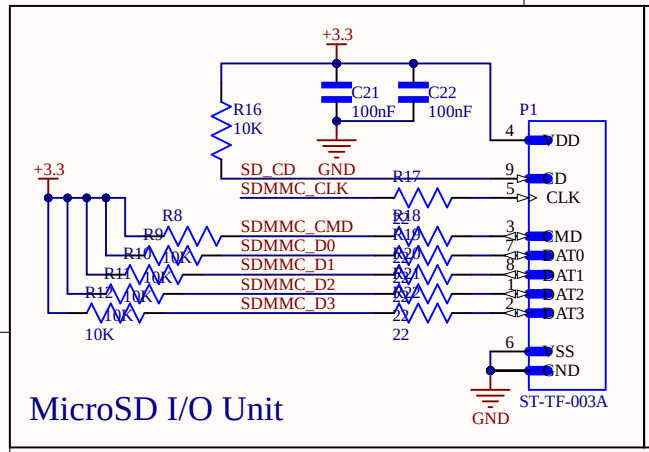
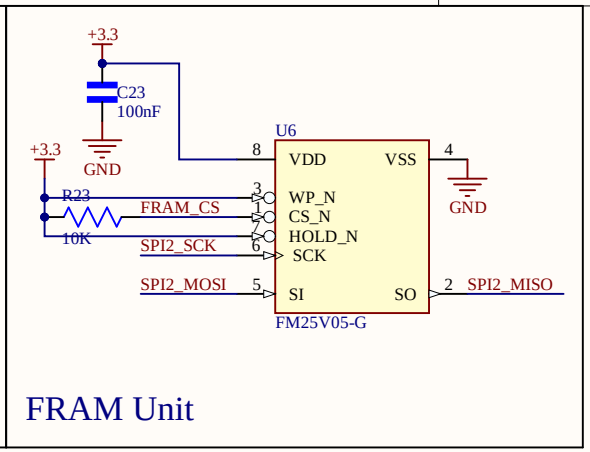
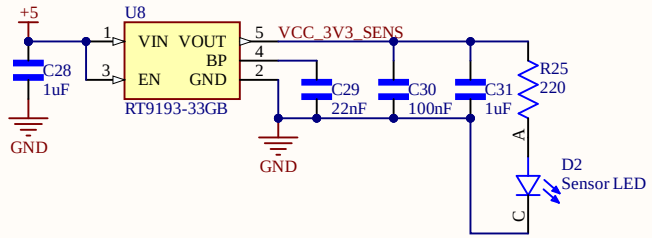

+3.3
S1
Safety ARM
D4
R33
220
SAFETY_SWITCH
Safety LED
R34
SAFETY_LED
D6
R35
FMU_ACT_LED
Kernel LED
D7
R36
FMU_ERR_LED
Error LED
U1
STM32H743VIT6
BOOT0
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2_C
PC3_C
PC4
PC5
PC6
PC7
PC8
PC9
PH0-OSC_IN
PH1-OSC_OUT
PC15-OSC32_OUT
PC14-OSC32_IN
VBAT
VDD
VDDA
VREF+
VCAP
NRST
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PC13
PC12
PC11
PC10
VSSA
VSS
VADC
RESET
CAN1_RX
CAN1_TX
SDMMC_CMD
FMU_ACT_LED
FMU_ERR_LED
USART2_TX
USART2_RX
USART3_TX
USART3_RX
AUX_PWM_1
AUX_PWM_2
AUX_PWM_3
AUX_PWM_4
BUZZER
UART8_RX
UART8_TX
FPGA_SPI_SCK
FPGA_CS
FPGA_SPI_MISO
FPGA_SPI_MOSI
UART7_RX
UART7_TX
MOTOR_1
MOTOR_2
MOTOR_3
MOTOR_4
PH0
PH1
PC15
PC14
SDMMC_CLK
SDMMC_D3
SDMMC_D2
UART4_TX
UART4_RX
BATT_VOLTAGE_SENS
BATT_CURRENT_SENS
SPI1_SCK
SPI1_MISO
SPI1_MOSI
IMU2_DRDY
VBUS
IMU1_DRDY
OTG_FS_P
OTG_FS_N
FMU-SWDIO
FMU-SWCLK
BARO1_CS
PB2-BOOT1
SPI3_SCK
SPI3_MISO
SPI3_MOSI
CAN2_TX
BARO2_CS
I2C1_SCL
I2C1_SDA
I2C2_SCL
I2C2_SDA
FRAM_CS
SPI2_SCK
SPI2_MISO
SPI2_MOSI
SAFETY_LED
SAFETY_SWITCH
IMU1_CS
IMU2_CS
FMU_USART6_TX
FMU_USART6_RX
SDMMC_D0
SDMMC_D1
R1
10K
R2
10K
R3
10K
R4
4K7
R5
4K7
R6
4K7
R7
4K7
C1
100nF
C2
100nF
C3
100nF
C4
100nF
C5
2.2uF
C6
2.2uF
C7
100nF
C8
100nF
C9
1uF
C10
100nF
C11
1uF
FB1
600
C12
10pF
C13
10pF
Y2
Crystal 32.768kHz
Y1
Crystal 24MHz
C39
8pF
C40
8pF
GND
U7
AMS1117-3.3
VIN
VOUT
ADJ/GND
C24
10uF
C25
100nF
C26
100nF
C27
10uF
R24
220
D1
MCU LED
MCU Power Unit
Title
Size
A4
Number
Revision
Date:
10/29/2026
Sheet of
File:
C:\Users\..\FMU_FMU.SchDoc
Drawn By:



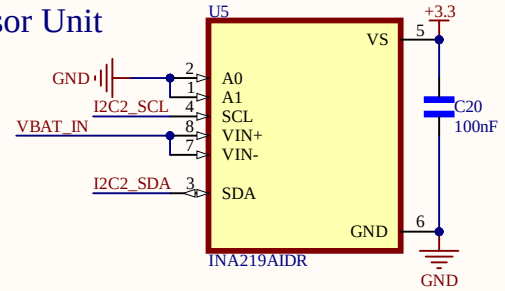


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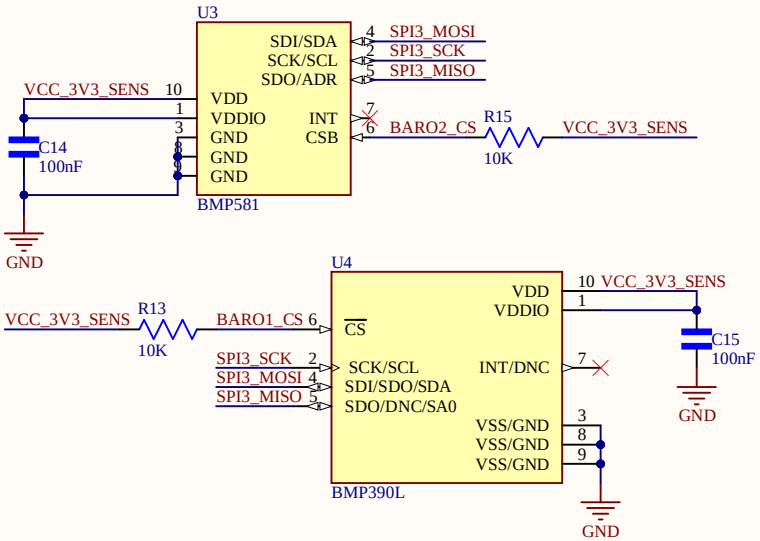
Sensor Power Unit



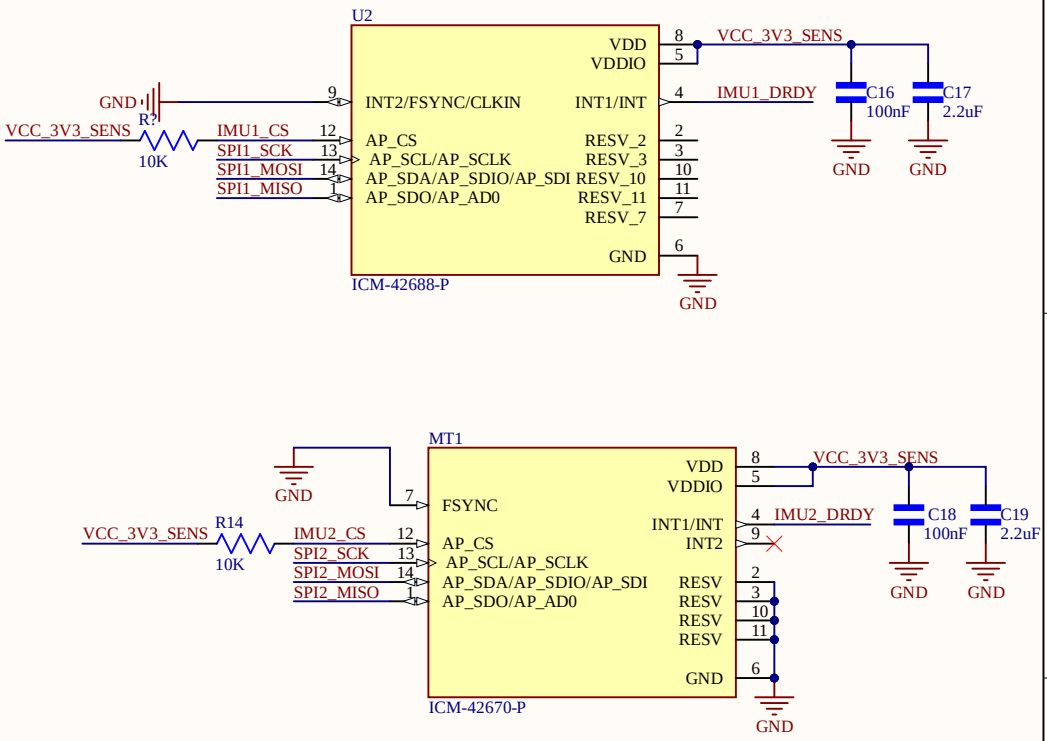
Voltage Sensor Unit



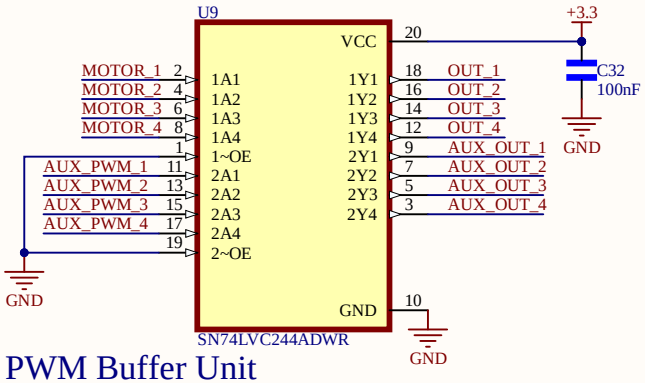
Barometer Unit



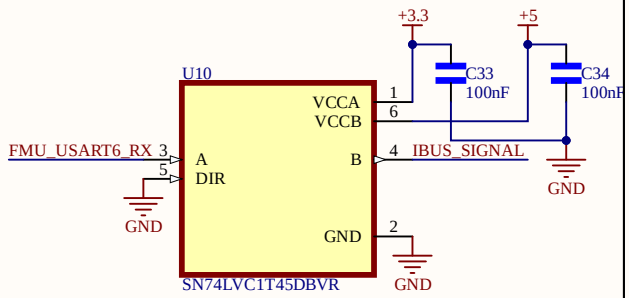
Inertial Measurement Unit



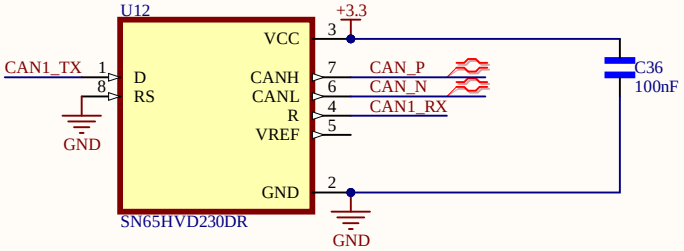
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Size A4	Number	Revision
Date: 10/29/2026		Sheet of
File: C:\Users\..\Sensor_FMU.SchDoc		Drawn By:



PWM Buffer Unit

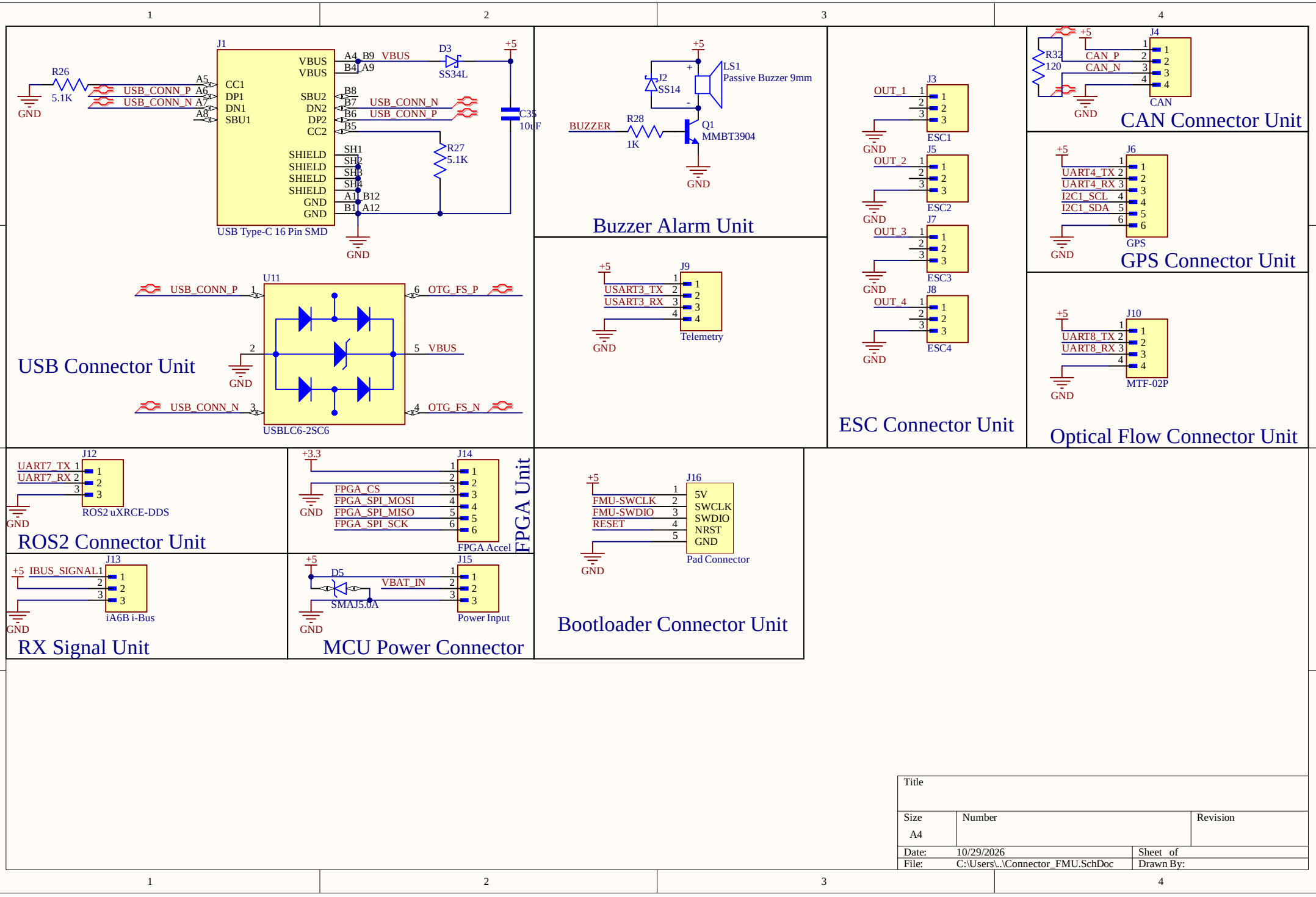


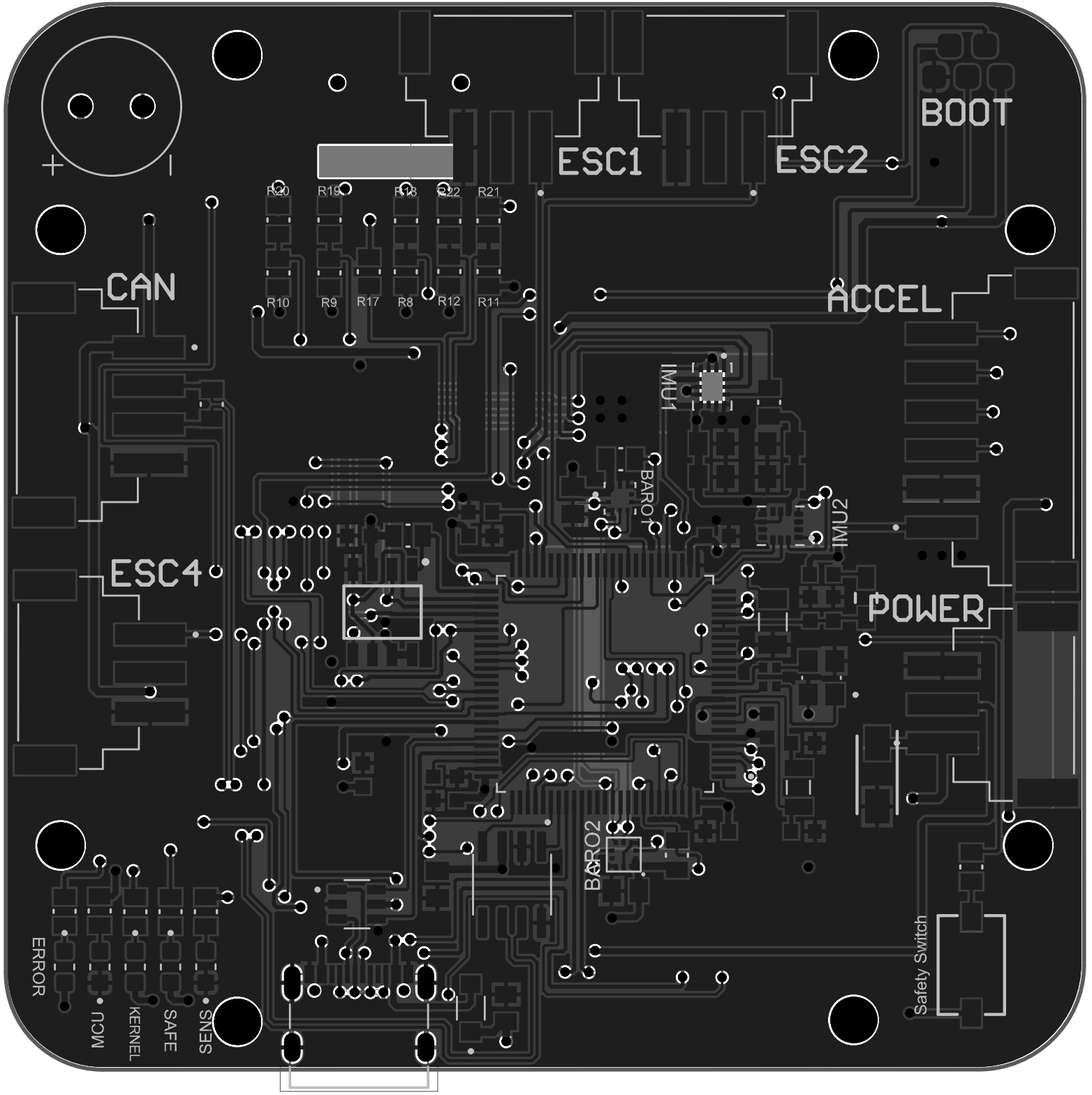
iBus Level Shifting Unit



CAN Bus Unit

Title		
Size: A4	Number	Revision
Date: 10/29/2026		Sheet of
File: C:\Users\..\PWM_FMU.SchDoc		Drawn By:


USB Connector Unit
USB Type-C 16 Pin SMD
Buzzer Alarm Unit
LS1
Passive Buzzer 9mm
MMBT3904
Telemetry
ESC Connector Unit
ESC1
ESC2
ESC3
ESC4
CAN Connector Unit
CAN
GPS Connector Unit
GPS
Optical Flow Connector Unit
MTF-02P
ROS2 Connector Unit
ROS2 uXRCE-DDS
FPGA Unit
FPGA Accel
RX Signal Unit
iA6B i-Bus
MCU Power Connector
Power Input
Bootloader Connector Unit
Pad Connector
USBLC6-2SC6
SMAJ5.0A
Title
Size
A4
Number
Revision
Date: 10/29/2026
File: C:\Users\..\Connector_FMU.SchDoc
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Drawn By:


BOOT
ESC1
ESC2
CAN
ACCEL
R20
R19
R18
R22
R21
R10
R9
R17
R8
R12
R11
IMU1
BARO1
IMU2
ESC4
POWER
BARO2
ERROR
MCU
KERNEL
SAFE
SENS
Safety Switch

Comment	Description	Designator	Footprint	LCSC	Quantity
100nF	100 nF ±10% 50V Ceramic Capacitor C0603 (1608 Metric)	C1, C2, C3, C4, C7, C11, C20, C21, C22, C23, C30	C_0603_1608Metric	C14663	11
0.1uF	0.1 µF ±10% 50V Ceramic Capacitor C0603 (1608 Metric)	C5, C9	C_0603_1608Metric	C14663	2
100nF	100 nF ±10% 50V Ceramic Capacitor C0603 (1608 Metric)	C8	C_0603_1608Metric	C14663	1
1uF	1 µF ±10% 50V Ceramic Capacitor C0603 (1608 Metric)	C6, C10, C26, C31	C_0603_1608Metric	C15849	4
10pF	10 pF ±10% 50V Ceramic Capacitor C0603 (1608 Metric)	C12, C13	C_0603_1608Metric	C1778	2
100nF	Multilayer Ceramic Capacitor 100 nF 50 V ±10% X7R 0603 (1608 Metric) Automotive AEC-Q200	C14, C15, C16, C18, C24, C25, C32, C33, C35, C36	C_0603_1608Metric	C14663	10
0.1uF	0.1 µF ±10% 25V Ceramic Capacitor C0603 (1608 Metric)	C17, C19	C_0603_1608Metric	C14663	2
22pF	22 pF ±5% 50V C0G/NP0 Multilayer Ceramic Capacitor 0603 (1608 Metric) Automotive AEC-Q200	C21, C27	C_0603_1608Metric	C1808	2
22nF	22 nF ±10% 50V Ceramic Capacitor C0603 (1608 Metric)	C28	C_0603_1608Metric	C14663	1
10nF	10 nF ±10% 16V Ceramic Capacitor C0603 (1608 Metric)	C29	C_0603_1608Metric	C15195	1
8pF	8 pF ±0.5pF 50V Ceramic Capacitor C0603 (1608 Metric)	C34, C37	C_0603_1608Metric	C1808	2
RC1 LED	Red 620nm LED Indication - Discrete 2.1V 0805 (2012 Metric)	D1	LED0_1206x1632mm_HandSolder	19801230379	1
Sensor LED	Red 620nm LED Indication - Discrete 2.1V 0805 (2012 Metric)	D2	LED0_1206x1632mm_HandSolder	19801230379	1
RS485	5V, 500, Planar Schottky	D3	SOD523	C181436	RS485	1
Safety LED	Red 620nm LED Indication - Discrete 2.1V 0805 (2012 Metric)	D4	LED0_1206x1632mm_HandSolder	19801230379	1
Watch.DA	5.5VDC, 600W, uni-directional, SMA pkg	D5	DO3041	C2290	Watch.DA	1
Relief LED	Red 620nm LED Indication - Discrete 2.1V 0805 (2012 Metric)	D6	LED0_1206x1632mm_HandSolder	19801230379	1
Error LED	Red 620nm LED Indication - Discrete 2.1V 0805 (2012 Metric)	D7	LED0_1206x1632mm_HandSolder	19801230379	1
MB	600V 800mA 4pin Through Hole Bridge Rectifier	D8	MB-DIA_0805_WD-3	MB-DIA_0805	1
USB Type-C 16 PCI	USB 2.0-only 16P Type-C Receptacle Connector	J1	USB_C_Receptacle_GCT_USB4125-GF-A_6Pin	C2765186	1
Mic1	Diode 500mW 5V 5% Unidirectional ESD Protection Diode 0402 (1005 Metric)	J2	Mic1	C2765186	1
RC1	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J3	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
LNA	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J4	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
RC2	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J5	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
DPA	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J6	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
RC3	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J7	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
RC4	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J8	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
Telemetry	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J9	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
MPP-02P	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J10	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
RC60 uBRC8-0201	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J12	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
RNS-I-Bus	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J13	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
RTSA Accel	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J14	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
Power Input	Connector Header Surface Mount, Right Angle 4-position 0.098" Pitch	J15	JST_SH_SM4B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	S3B-XH-SM3-TB	1
Test connector	Test connector	J16	Test connector PCB	Test connector	1
Reboot Button Mech	SLIDE THROUGHBOARD SPST 100MA 20DRAW	J18	SSCH_AT-0627-TT-2-H	AT-0627-TT-2-H	1
CW_42639-P	CW_42639-P	MT1	SSCH_CW-42637-P	CW_42639-P	1
SY-TP-S01A	Micro-SD Card Socket	P1	SUNTOUCH_SY-TP-S01A	SY-TP-S01A	1
MAX97356A	BAT, 400~25, 400V, 1000mA, 50Pk, 0.2000, 0.0000 R	Q1	TYPHU_MAX97356A	MAX97356A	1
10K	10 kOhms ±1% 1/10W 1/16W Chip Resistor 0603 (1608 Metric) Moisture Sensitive Thick Film	R1, R2, R3, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24	R_0603_1608Metric	R10603	24
887	1.5 kOhms ±1% 1/10W 1/16W Chip Resistor 0603 (1608 Metric) Moisture Sensitive Thick Film	R4, R5, R6, R7	R_0603_1608Metric	R10603	4
1K	1 kOhms ±1% 1/10W 1/16W Chip Resistor 0603 (1608 Metric) Moisture Sensitive Thick Film	R17, R18, R19, R20, R21, R22	R_0603_1608Metric	R10603	6
330	330 Ohms ±1% 1/10W 1/16W Chip Resistor 0603 (1608 Metric) Moisture Sensitive Thick Film	R23, R25, R33, R34, R35, R36	R_0603_1608Metric	R10603	6
4.1K	4.1 kOhms ±1% 1/10W 1/16W Chip Resistor 0603 (1608 Metric) Moisture Sensitive Thick Film	R26, R27	R_0603_1608Metric	R10603	2
1K	1 kOhms ±1% 1/10W 1/16W Chip Resistor 0603 (1608 Metric) Moisture Sensitive Thick Film	R28	R_0603_1608Metric	R10603	1
10K	10 kOhms ±1% 1/10W 1/16W Chip Resistor 0603 (1608 Metric) Moisture Sensitive Thick Film	R32	R_0603_1608Metric	R10603	1
Safety ARM	SLPT-L1004 IPRT terminal type, SMT, Height: 2.5mm, Black Housing, Gold Plated Contacts, 100pF, TMR	R1	1337S1SR616	1337S1SR616	1
HYM2024T20T6	20MHz/T Crystal/±7~50PPM/Industrial/-40~+85℃/3.2mm*2.5mm*0.7mm/4P SMD	R5	QPRHP1400S1400S1241-1000	HYM2024T20T6	1
IGM -02688-P	Accelerometer, Gyroscope, 6 Axis	R7	PGPHW3P3002S050707-MR	IGM -02688-P	1
MAP581	Very small, low power, low noise 28-bit analog pressure sensor. This digital, high performance sensor is ideally suited for a wide range of altitude sensing applications including outstanding design flexibility	R8	MAP581	MAP581	1
MAP3581	Pressure Sensor (L3MPG - 3A 13PG 15MPA ~ 15MPA) Absolute ~ 10 MPA	R9	I2C5_MAP3581	MAP3581	1
RA0C564A0R	2M ± 1/2 kΩ ±1% 1/4W Thick Film	R10	I2C12 TP0R0K176-BN	RA0C564A0R	1
Backbyte-G	1 kOhm 5kΩ ± 4Ω Metal Alloy	R11	I2C12 TP4R0K173-BN	Backbyte-G	1
AMS1117-3.3	Linear Voltage Regulator IC Positive Fixed 1 Output 1A SOT-223-3	R12	I2C12 TP0R0K180-BN	AMS1117-3.3	1
RTR303-3003E	Linear Voltage Regulator IC Positive Fixed 1 Output 300mA SOT-23-3	R13	I2C12 TP0R0K176-BN	RTR303-3003E	1
RA7SJC1C1G0CNR	0 Ohms ±5% 0.5 W Surface Mount Resistor 2010 (5025 Metric)	R14	I2C12 TP1G0K200-BN	RA7SJC1C1G0CNR	1
RA7SJC1T1B0BVR	Single-Bit Dual-Supply Bus Transceiver with Configurable Voltage Level Shifting and 3-State Outputs	R15	I2C TP0R0K181-BN	RA7SJC1T1B0BVR	1
10MLCS-15CX	15 V Clamp 54 W Bi-Directional Dual TVS Diode Surface Mount - SMA-J5.0A	R16	I2C TP0R0K176-BN	10MLCS-15CX	1
RA6MHG1G0DCR	0 Ω ±5% 1W Automotive Grade Thick Film	R17	I2C12 TP0R0K176-BN	RA6MHG1G0DCR	1
Crystal 100MHz	100 MHz ±10ppm Crystal 20pF 60 Ohms 4-SMD, No Lead	T1	XTAL_ABM3B-100.000MHZ-10-1-U-20PF-32-D-TR	ABM3B-100.000MHZ-10-1-U-20PF-32-D-TR	1
Crystal 32.768kHz	LOW FREQUENCY CRYSTAL	T2	XTAL_12.5Y-T1A-32.768KHZ-7PF-±20PPM-TA-QC	12.5Y-T1A-32.768KHZ-7PF-±20PPM-TA-QC	1